Carleton University

Department of Systems and Computer Engineering

Normalization Evidence

Third Normal Form (3NF) Compliance

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1. Overview and 3NF Requirements

1.1 Purpose of This Document

This document provides formal evidence that the Health and Fitness Club Management System database schema satisfies Third Normal Form (3NF) requirements. Each of the 15 tables in the schema is analysed individually, with all attributes, functional dependencies, and compliance status documented in full.

1.2 3NF Requirements Recap

A relation is in Third Normal Form if and only if it satisfies both of the following conditions:

(1) It is in Second Normal Form (2NF) — meaning it is already in First Normal Form and every non-prime attribute is fully functionally dependent on the entire primary key (no partial dependencies).

(2) No transitive dependencies exist — meaning no non-prime attribute depends on another non-prime attribute. Every non-prime attribute must depend directly and solely on the primary key.

1.3 Schema at a Glance

Table	PK	Foreign Keys	Total Attributes
Users	id (UUID)	—	6
Members	id (UUID)	id → users.id	8
Trainers	id (UUID)	id → users.id	5
Admin_Staff	id (UUID)	id → users.id	5
Rooms	id (UUID)	—	5
Equipments	id (UUID)	room_id → rooms.id	8
Subscriptions	id (UUID)	—	5
Member_Subscriptions	id (UUID)	member_id, subscription_id	8

Table	PK	Foreign Keys	Total Attributes
Fitness_Goals	id (UUID)	member_id → members.id	6
Health_Metrics	id (UUID)	member_id → members.id	6
Trainer_Availability	id (UUID)	trainer_id → trainers.id	6
Classes	id (UUID)	trainer_id, room_id	10
Enrollments	id (UUID)	member_id, class_id	4
Training_Sessions	id (UUID)	trainer_id, member_id, room_id	10
Payments	id (UUID)	member_id, subscription_id	9

2. Schema Analysis — All 15 Tables

Each table is analysed below. The attribute analysis table lists every column, classifies it as prime or non-prime, states its functional dependency, and provides contextual notes. A compliance badge confirms 3NF satisfaction.

1. Users

Primary Key id (UUID)

Attribute Analysis

Attribute	Type	Dependency	Notes
id	Prime	Primary key	UUID, base authentication
email	Non-prime	Depends on id	Unique per user
password	Non-prime	Depends on id	Stored as bcrypt hash
role	Non-prime	Depends on id	ENUM: member, trainer, admin
created_at	Non-prime	Depends on id	Auto-set on insert
updated_at	Non-prime	Depends on id	Maintained by trigger

Analysis Notes

- All non-prime attributes depend directly and solely on the primary key.
- No transitive dependencies: role does not determine any other attribute.
- Single-attribute PK eliminates any possibility of partial dependencies (2NF trivially satisfied).

3NF Status: SATISFIED

2. Members

Primary Key id (UUID)

Foreign Key id → users.id (1:1 inheritance)

Attribute Analysis

Attribute	Type	Dependency	Notes
id	Prime	Primary key / FK	Shares PK with Users (role separation)
full_name	Non-prime	Depends on id	Member's display name
date_of_birth	Non-prime	Depends on id	Not derived from any other attribute
gender	Non-prime	Depends on id	No dependency on other columns
phone	Non-prime	Depends on id	Not derivable from email or name
created_at	Non-prime	Depends on id	Auto-set on insert
updated_at	Non-prime	Depends on id	Maintained by trigger
deleted_at	Non-prime	Depends on id	Soft delete timestamp

Analysis Notes

- Storing profile data in Members rather than Users prevents transitive dependency Members.phone → Members.email → Users.id.
- The 1:1 FK to Users is an integrity constraint, not a transitive dependency.

3NF Status: SATISFIED

3. Trainers

Primary Key id (UUID)

Foreign Key id → users.id (1:1 inheritance)

Attribute Analysis

Attribute	Type	Dependency	Notes
id	Prime	Primary key / FK	Shared PK with Users
full_name	Non-prime	Depends on id	Trainer display name
created_at	Non-prime	Depends on id	Auto-set on insert

Attribute	Type	Dependency	Notes
updated_at	Non-prime	Depends on id	Maintained by trigger
deleted_at	Non-prime	Depends on id	Soft delete timestamp

Analysis Notes

- Minimal attribute set; all attributes depend directly on the PK.
- Trainer-specific attributes (availability, sessions) are stored in separate tables.

3NF Status: SATISFIED

4. Admin_Staff

Primary Key id (UUID)

Foreign Key id → users.id (1:1 inheritance)

Attribute Analysis

Attribute	Type	Dependency	Notes
id	Prime	Primary key / FK	Shared PK with Users
full_name	Non-prime	Depends on id	Admin display name
created_at	Non-prime	Depends on id	Auto-set on insert
updated_at	Non-prime	Depends on id	Maintained by trigger
deleted_at	Non-prime	Depends on id	Soft delete timestamp

Analysis Notes

- Identical structure to Trainers; role separation maintained via Users.role ENUM.

3NF Status: SATISFIED

5. Rooms

Primary Key id (UUID)

Attribute Analysis

Attribute	Type	Dependency	Notes
id	Prime	Primary key	UUID
name	Non-prime	Depends on id	Unique room label
capacity	Non-prime	Depends on id	Not derived from name
created_at	Non-prime	Depends on id	Auto-set on insert
updated_at	Non-prime	Depends on id	Maintained by trigger

Analysis Notes

- capacity does not depend on name; two rooms can share a capacity value independently.
- No transitive dependencies present.

3NF Status: SATISFIED

6. Equipments

Primary Key id (UUID)

Foreign Key room_id → rooms.id

Attribute Analysis

Attribute	Type	Dependency	Notes
id	Prime	Primary key	UUID
room_id	Non-prime	Depends on id	FK reference — not a transitive dependency
equipment_name	Non-prime	Depends on id	Name of this specific unit
status	Non-prime	Depends on id	ENUM: operational, under_repair, out_of_service

Attribute	Type	Dependency	Notes
maintenance_notes	Non-prime	Depends on id	Free-text notes for this unit
created_at	Non-prime	Depends on id	Auto-set on insert
updated_at	Non-prime	Depends on id	Maintained by trigger
deleted_at	Non-prime	Depends on id	Soft delete timestamp

Analysis Notes

- room_id is a FK referencing rooms.id. It records which room the equipment belongs to — it is not a derived or transitively dependent value.
- maintenance_notes depends on the specific equipment record (id), not on status.

3NF Status: SATISFIED

7. Subscriptions

Primary Key id (UUID)

Attribute Analysis

Attribute	Type	Dependency	Notes
id	Prime	Primary key	UUID
plan	Non-prime	Depends on id	Plan name/tier
fee	Non-prime	Depends on id	Fee is set per record, not derived from plan name
created_at	Non-prime	Depends on id	Auto-set on insert
updated_at	Non-prime	Depends on id	Maintained by trigger

Analysis Notes

- fee does not depend on plan; different plans may independently have the same fee value without creating a functional dependency.

3NF Status: SATISFIED

8. Member_Subscriptions

Primary Key id (UUID)

Foreign Key member_id → members.id

subscription_id → subscriptions.id

Attribute Analysis

Attribute	Type	Dependency	Notes
id	Prime	Primary key	UUID
member_id	Non-prime	Depends on id	FK to Members — not transitive
subscription_id	Non-prime	Depends on id	FK to Subscriptions — not transitive
start_date	Non-prime	Depends on id	Specific to this assignment record
end_date	Non-prime	Depends on id	Specific to this assignment record
status	Non-prime	Depends on id	ENUM: active, expired, cancelled
created_at	Non-prime	Depends on id	Auto-set on insert
updated_at	Non-prime	Depends on id	Maintained by trigger

Analysis Notes

- Separating subscription assignments from the Members table prevents storing plan fee data redundantly per member row.
- Both FK columns reference other tables' PKs; they are relationship pointers, not transitively dependent attributes.

3NF Status: SATISFIED

9. Fitness_Goals

Primary Key id (UUID)

Foreign Key member_id → members.id

Attribute Analysis

Attribute	Type	Dependency	Notes
id	Prime	Primary key	UUID
member_id	Non-prime	Depends on id	FK to Members
description	Non-prime	Depends on id	Goal description text
target_value	Non-prime	Depends on id	Numeric target (e.g. 70 kg)
created_at	Non-prime	Depends on id	Auto-set on insert
updated_at	Non-prime	Depends on id	Maintained by trigger

Analysis Notes

- Storing goals separately from Members prevents embedding multi-valued health data in the member profile and avoids update anomalies.

3NF Status: SATISFIED

10. Health_Metrics

Primary Key id (UUID)

Foreign Key member_id → members.id

Attribute Analysis

Attribute	Type	Dependency	Notes
id	Prime	Primary key	UUID
member_id	Non-prime	Depends on id	FK to Members
metric_type	Non-prime	Depends on id	E.g. weight, heart_rate, BMI
metric_value	Non-prime	Depends on id	Numeric reading
recorded_at	Non-prime	Depends on id	Timestamp of measurement
created_at	Non-prime	Depends on id	Auto-set on insert

Analysis Notes

- metric_value does not depend on metric_type; the type label is descriptive, not a functional determinant of the value.
- One row per measurement enables time-series health tracking without column explosion.

3NF Status: SATISFIED

11. Trainer_Availability

Primary Key id (UUID)

Foreign Key trainer_id → trainers.id

Attribute Analysis

Attribute	Type	Dependency	Notes
id	Prime	Primary key	UUID
trainer_id	Non-prime	Depends on id	FK to Trainers
available_date	Non-prime	Depends on id	Calendar date of availability
start_at	Non-prime	Depends on id	Availability window start time
end_at	Non-prime	Depends on id	Availability window end time
created_at	Non-prime	Depends on id	Auto-set on insert

Analysis Notes

- start_at and end_at are specific to this availability slot; they do not depend on each other or on available_date — each combination is stored as a separate record.

3NF Status: SATISFIED

12. Classes

Primary Key id (UUID)

Foreign Key trainer_id → trainers.id

room_id → rooms.id

Attribute Analysis

Attribute	Type	Dependency	Notes
id	Prime	Primary key	UUID
name	Non-prime	Depends on id	Class name / label

Attribute	Type	Dependency	Notes
trainer_id	Non-prime	Depends on id	FK to Trainers
room_id	Non-prime	Depends on id	FK to Rooms
class_date	Non-prime	Depends on id	Scheduled date
start_time	Non-prime	Depends on id	Class start time
end_time	Non-prime	Depends on id	Class end time
max_capacity	Non-prime	Depends on id	Capacity for this class instance
created_at	Non-prime	Depends on id	Auto-set on insert
deleted_at	Non-prime	Depends on id	Soft delete timestamp

Analysis Notes

- max_capacity is an attribute of the specific class record, not of the room or trainer.
- No attribute depends on another non-prime attribute; all depend solely on class id.

3NF Status: SATISFIED

13. Enrollments (Junction Table)

Primary Key id (UUID)

Foreign Key member_id → members.id
class_id → classes.id

Attribute Analysis

Attribute	Type	Dependency	Notes
id	Prime	Primary key	UUID surrogate key
member_id	Non-prime	Depends on id	FK to Members
class_id	Non-prime	Depends on id	FK to Classes
registered_at	Non-prime	Depends on id	Timestamp of enrolment

Analysis Notes

- Unique constraint on (member_id, class_id) prevents duplicate enrolments.
- The junction table resolves the M:N relationship between Members and Classes without any transitive dependencies.

3NF Status: SATISFIED

14. Training_Sessions

Primary Key id (UUID)

Foreign Key trainer_id → trainers.id

member_id → members.id

room_id → rooms.id

Attribute Analysis

Attribute	Type	Dependency	Notes
id	Prime	Primary key	UUID
trainer_id	Non-prime	Depends on id	FK to Trainers
member_id	Non-prime	Depends on id	FK to Members
room_id	Non-prime	Depends on id	FK to Rooms
session_date	Non-prime	Depends on id	Scheduled session date
start_time	Non-prime	Depends on id	Session start time
end_time	Non-prime	Depends on id	Session end time
status	Non-prime	Depends on id	ENUM: scheduled, completed, cancelled
created_at	Non-prime	Depends on id	Auto-set on insert
updated_at	Non-prime	Depends on id	Maintained by trigger

Analysis Notes

- Functions as both a data table and a junction table resolving the M:N relationship between Members and Trainers.
- status does not depend on session_date or times; it is independently updatable.
- The overlap-prevention trigger enforces scheduling constraints without violating 3NF.

3NF Status: SATISFIED

15. Payments

Primary Key	id (UUID)
Foreign Key	member_id → members.id
	subscription_id → member_subscriptions.id

Attribute Analysis

Attribute	Type	Dependency	Notes
id	Prime	Primary key	UUID
member_id	Non-prime	Depends on id	FK to Members
subscription_id	Non-prime	Depends on id	FK to Member_Subscriptions
amount	Non-prime	Depends on id	Actual amount charged (not derived from plan)
paid_at	Non-prime	Depends on id	Payment timestamp
payment_method	Non-prime	Depends on id	E.g. credit_card, paypal
status	Non-prime	Depends on id	ENUM: pending, completed, failed, refunded
deleted_at	Non-prime	Depends on id	Soft delete timestamp
created_at	Non-prime	Depends on id	Auto-set on insert

Analysis Notes

- amount is stored per payment record and is not derived from the subscription fee; this allows for discounts or partial payments without schema changes.
- payment_method does not depend on amount or status; all three depend solely on id.

3NF Status: SATISFIED

3. Key Design Decisions Supporting 3NF

3.1 Elimination of Transitive Dependencies

(1) User Role Separation. Rather than embedding role-specific attributes directly in the Users table, three separate tables (Members, Trainers, Admin_Staff) are created with 1:1 FK relationships to Users. This eliminates scenarios such as Users.phone → Users.email → Users.id which would violate 3NF.

(2) Health Data Separation. Health metrics and fitness goals are stored in dedicated tables (Health_Metrics, Fitness_Goals) rather than as columns on the Members table. Embedding them would force nullable or repeated columns and could create transitive dependencies over time as new metric types are added.

(3) Training Data Separation. Training sessions and class enrolments are stored in their own tables and linked via foreign keys, keeping member profile data strictly within the Members table.

(4) Subscription Assignment Separation. Plan definitions (name, fee) are stored in Subscriptions, while a member's plan assignment is recorded in Member_Subscriptions. Storing the fee inside Member_Subscriptions would create a transitive dependency: fee → subscription_id → id.

3.2 Proper Foreign Key Usage

All foreign keys in the schema reference primary keys of other tables. They are used solely to establish relationships between entities and do not store derived, calculated, or duplicated data. A foreign key column records a reference pointer; it does not create a transitive dependency because the referenced value is the primary key of an independent relation.

3.3 Atomic Attributes

Every attribute across all 15 tables holds a single, indivisible value. No column contains comma-delimited lists, JSON arrays, or other composite structures that would violate the First Normal Form foundation required for 3NF compliance. For example, a member with multiple phone numbers would require a separate PhoneNumbers table — though such a requirement is outside the current project scope.

3.4 Single-Purpose Tables

Each table is designed around a single, clearly defined entity or relationship. No table conflates two or more concerns. This principle directly supports 3NF by ensuring that all non-prime attributes within a table are semantically related to the same entity, making transitive dependencies structurally improbable.

3.5 No Stored Derived Data

No attribute in the schema stores a value that is calculable from other attributes within the same table. For example, a member's age is not stored because it can be derived from date_of_birth. Storing derived data would introduce update anomalies and could create functional dependencies between non-prime attributes.

Design Pattern	3NF Benefit
Users / Members / Trainers split	Eliminates transitive role-attribute dependencies
Health_Metrics as separate table	Prevents member profile update anomalies
Subscriptions / Member_Subscriptions	Avoids fee duplication per member row
FK-only relationships	No derived data stored; references only
Atomic column values	Satisfies 1NF → foundation for 3NF
Single-purpose tables	All non-prime attributes semantically cohesive
No calculated columns	Removes inter-attribute functional dependencies

4. Summary and Conclusion

4.1 Compliance Summary

The table below consolidates the 3NF compliance status for all 15 tables in the Health and Fitness Club Management System schema.

Table	1NF	2NF	3NF	Transitive Deps?
Users				None
Members				None
Trainers				None
Admin_Staff				None
Rooms				None
Equipments				None
Subscriptions				None
Member_Subscriptions				None
Fitness_Goals				None
Health_Metrics				None
Trainer_Availability				None
Classes				None
Enrollments				None
Training_Sessions				None
Payments				None

4.2 Conclusion

The Health and Fitness Club Management System database schema fully and demonstrably satisfies Third Normal Form across all 15 tables. The evidence presented in this document confirms that:

- All tables satisfy First Normal Form — every attribute is atomic and a primary key is defined for every table.
- All tables satisfy Second Normal Form — single-attribute UUID primary keys make partial dependencies structurally impossible throughout the schema.
- All tables satisfy Third Normal Form — no non-prime attribute depends on another non-prime attribute; all dependencies lead directly to the primary key.
- No transitive dependencies exist anywhere in the schema, confirmed through the explicit analysis of every attribute in every table.
- Data integrity is maintained through appropriate use of foreign keys, constraint declarations, and purposeful table separation.

Overall Schema 3NF Compliance: FULLY SATISFIED — All 15 Tables

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